Q1.

(a) Sunil picked yellow, red and purple primula flowers from his garden.

He dipped the different flower petals into water and into two different solutions.
The pH of one solution was 1 and the pH of the other was 10.
The table shows the results.

colour of flower petals	in solutions of pH 1	in water pH 7	in solution of pH 10
yellow	stayed yellow	stayed yellow	stayed yellow
red	stayed red	stayed red	turned green
purple	turned pink	stayed purple	turned blue

Which colour of flower petal would be most useful to make an indicator for **both** acids **and** alkalis? Explain your answer.

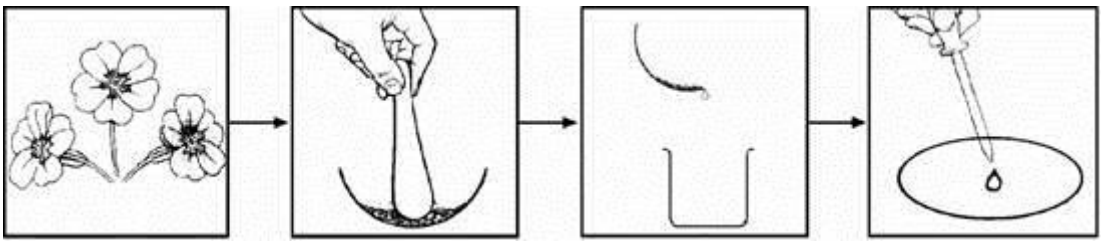
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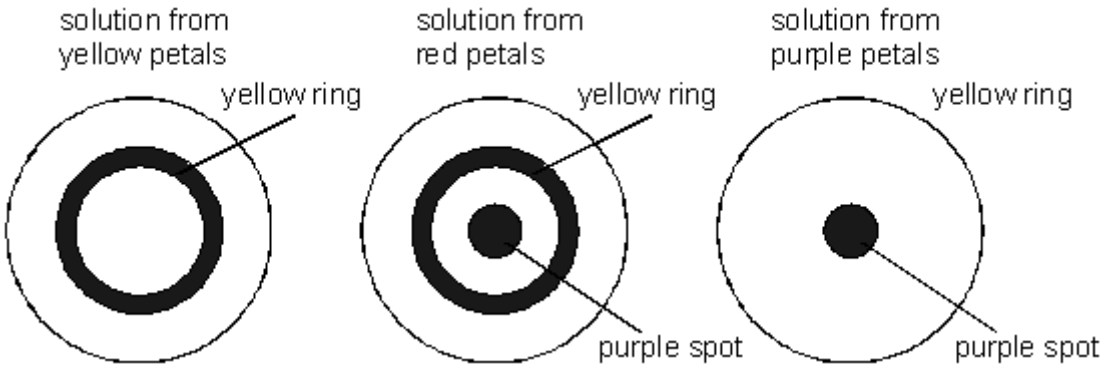
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2 marks

Sunil crushed petals from each flower separately in some liquid and poured off the coloured solutions. Then he put drops of each coloured solution into the middle of different pieces of filter paper.



The solutions spread out on the filter paper. The diagrams show his results.



(b) What is the name of this method of investigating coloured substances?

.....

1 mark

(c) Sunil made notes on his experiment. Some words are missing.
Complete the sentences.

When I crushed a flower in a liquid it produced a coloured solution.

This is because a coloured substance had .....…….. in the liquid. This shows that the liquid is a ..... for these coloured substances.

My experiment shows that one of the flowers probably contained two coloured substances. This was the .....…….. flower.

3 marks
Maximum 6 marks

Q1.

(a) purple

1

any **one** from

- it changes colour in both acids and alkalis

  *do **not** accept 'it changes colour in acids'*
  ***or** 'it changes colour in alkalis'*

- it goes pink in acid or pH1 and blue in alkaline or pH10

1

(b) chromatography

1

(c) dissolved

answers must be in the correct order
*do **not** accept 'made a solution'*

1

solvent

1

red

accept 'second'

1

[6]